



**TEMASEK
JUNIOR COLLEGE**

CHEMISTRY

8872/01

Paper 1 Multiple Choice

Friday

23rd SEPTEMBER 2011

50 minutes

Additional materials: Data Booklet
 Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, index number and CG on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

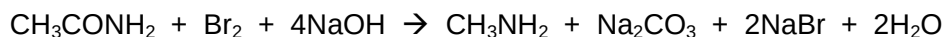
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers, **A, B, C and D**. Choose the **one** you consider to be correct.

- 1** The Hofmann degradation is an example of a step-down reaction between an amide and a mixture of bromine and sodium hydroxide solution to form amines.

The following shows an equation for the preparation of methylamine:



What is the mass of sodium hydroxide required for the reaction, given that 16.5 g of methylamine is produced?

[A_r of C = 12.0; H = 1.0; N = 14.0; O = 16.0; Na = 23.0]

- A** 5.3 g
- B** 21.3 g
- C** 66.2 g
- D** 85.2 g
- 2** Use of the *Data Booklet* is relevant to this question.

The combustion of fossil fuels is a major source of increasing atmospheric carbon dioxide, with a consequential rise in global warming. Other significant contributions to carbon dioxide levels comes from the thermal decomposition of limestone (calcium carbonate) and the manufacture of cement and lime for agricultural purposes.

Cement works roast 1000 million tonnes of limestone per year and a further 200 million tonnes is roasted in kilns to make lime.

What is the total annual mass output of carbon dioxide (in million tonnes) from these two processes?

- A** 440
- B** 528
- C** 660
- D** 880

- 3 In an experiment, 0.0050 mol of a metallic salt reacted with 0.0025 mol of aqueous sodium sulfate(IV).

The half-equation for oxidation of sulfate(IV) ion is shown below.



If the original oxidation number of the metal in the salt was +3, what would be the new oxidation number of the metal?

- A** 0 **B** +1 **C** +2 **D** +4
- 4 Which of the following is correct about isotopes of an element?
- A** They have the same mass number.
B They have more protons than neutrons.
C They have the same electronic configuration.
D They have the same number of protons and neutrons.
- 5 Which of the following species have three unpaired electrons?
- A** Al
B Ti^{3+}
C Zn^{2+}
D Cr^{3+}
- 6 Which of the following has the largest bond angle?
- A** IF_2^-
B IF_4^-
C PH_3
D SiCl_4

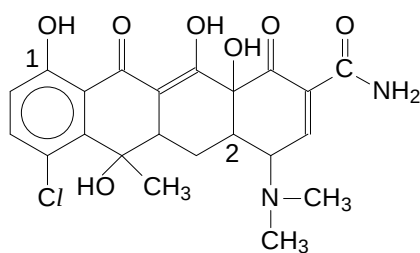
7 Which substance only exhibits ionic bonding?

- A** H_2CO_3 **B** BaSO_4 **C** ZnF_2 **D** NH_4Cl

8 Which of the following molecules will not have hydrogen bonding with another of its own molecules?

- A** $\text{CH}_3\text{CO}_2\text{H}$ **B** CH_3CHO **C** CH_3NH_2 **D** CH_3OH

9 Aureomycin is a powerful antibiotic.



Aureomycin

Which of the following is correct about the hybridisation of the labelled carbon atoms and its corresponding shape?

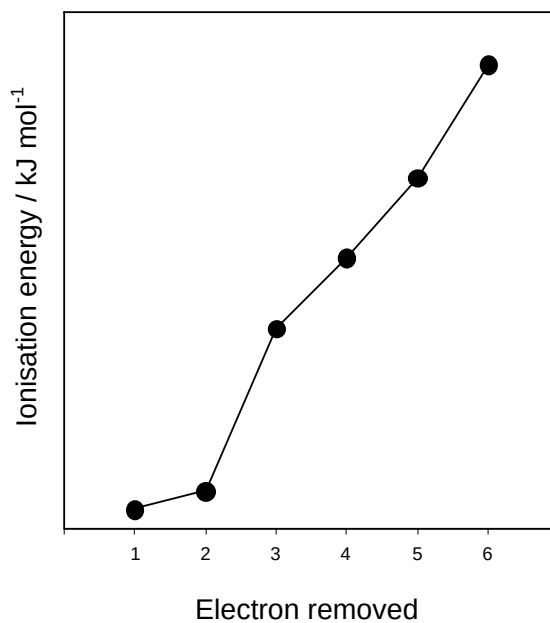
	Carbon 1	Carbon 2
A	sp^2 , trigonal planar	sp^3 , trigonal planar
B	sp^2 , trigonal planar	sp^3 , tetrahedral
C	sp^3 , trigonal planar	sp^2 , tetrahedral
D	sp^3 , tetrahedral	sp^3 , tetrahedral

10 Which of the following statements regarding elements in Period 3 (i.e. sodium to chlorine) and their compounds is **incorrect** as atomic number increases?

- A** The oxides become increasingly acidic.
B The chlorides become increasingly covalent.
C The melting points of the elements generally increase.
D The atomic radii of the elements generally decrease.

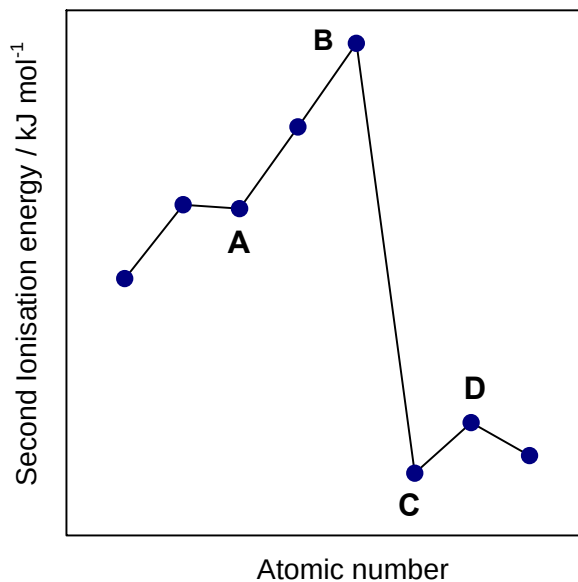
- 11 **Figure 1** shows the first six ionisation energies of an element **Q**.

Figure 1



- Figure 2** shows the second ionisation energies of eight consecutive elements (including **Q**).

Figure 2



With reference to **Figure 2**, which of the options, **A**, **B**, **C** or **D**, corresponds to the second ionisation energy of element **Q**?

- 12** Element **R** is in Period 3. It is burnt in air to produce a yellow flame and forms a white solid. The white solid is then dissolved in an aqueous solution containing universal indicator.

Which of the following can be deduced from the above observation?

	Identity of Element R	Colour of aqueous solution
A	Sodium	Blue
B	Magnesium	Red
C	Phosphorus	Blue
D	Sulfur	Red

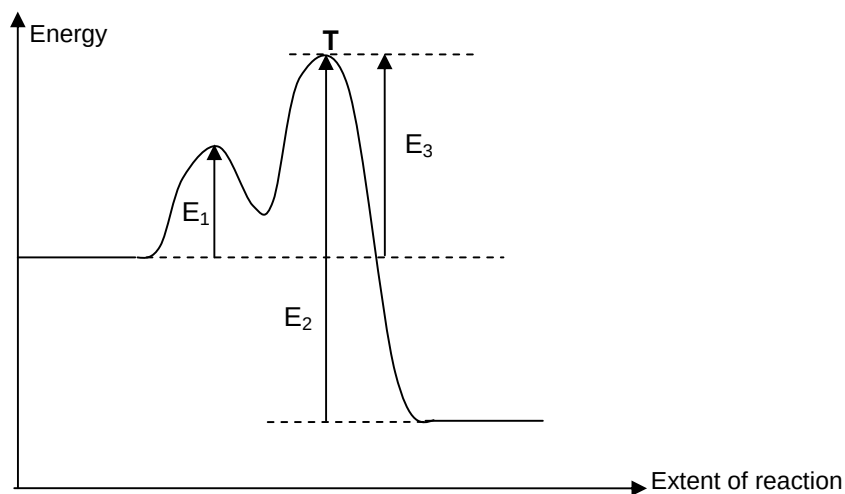
- 13** The lattice energy of two compounds is shown below.

compound	Lattice energy / kJ mol^{-1}
Sodium chloride	-766
Magnesium oxide	-3889

Which statement best explains the difference in the numerical value of the lattice energy?

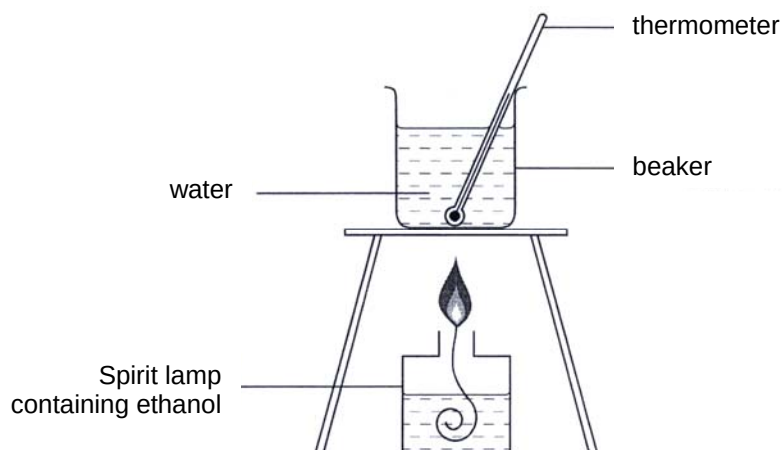
- A** The charge density of magnesium ion is greater than that of sodium ion.
- B** More energy is required to form the ionic bonds in magnesium oxide.
- C** The atomic radius of magnesium is larger than that of sodium.
- D** The charges of both the magnesium ion and oxide ion are larger.

14



Which of the following statements is true about the energy profile above?

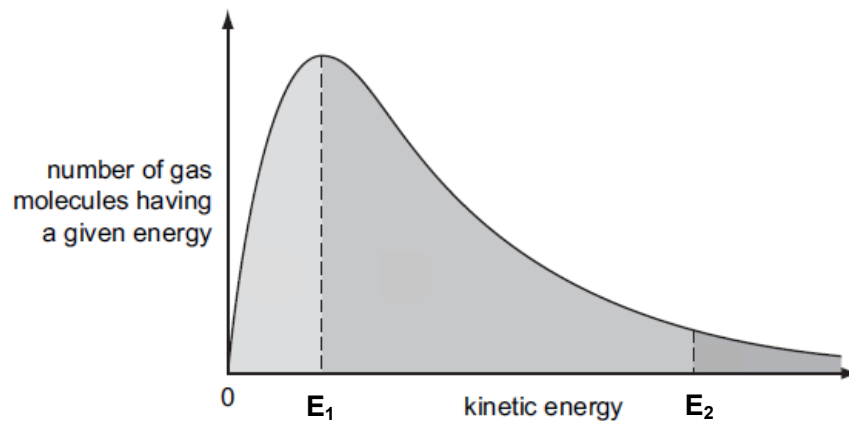
- A** T is the intermediate formed.
 - B** Activation energy for the first step is $E_3 - E_1$.
 - C** The enthalpy change of the reaction is $E_3 - E_2$.
 - D** E_2 corresponds to the activation energy for the reaction in the second step.
- 15** A student set up the apparatus under room conditions as shown below to heat a beaker containing 400 g of water using 2.50 g of ethanol.



Given that the standard enthalpy change of combustion of ethanol is $-1370 \text{ kJ mol}^{-1}$ and the efficiency was found to be only 70 %, determine the final temperature recorded.

- | | |
|------------------|------------------|
| A 31.2 °C | C 44.5 °C |
| B 56.2 °C | D 69.5 °C |

- 16** The Boltzmann distribution shows the number of molecules having a particular kinetic energy at constant temperature.



Which of the following describe the change in the number of molecules when temperature is increased by 10 °C?

	Number of molecules with energy		
	$\leq E_1$	Between E_1 and E_2	$\geq E_2$
A	Decrease	Increase	Decrease
B	Decrease	Increase	Increase
C	Increase	Decrease	Decrease
D	Increase	Increase	decrease

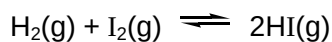
- 17** Lime can be used to neutralise acidity of lake water caused by acid rain. The pH value of the lake water rises from 4 to 7.

What is the change in concentration of H^+ ions in the lake?

- A** An increase by a factor of 3.
- B** An increase by a factor of 1000.
- C** A decrease by a factor of 3.
- D** A decrease by a factor of 1000.

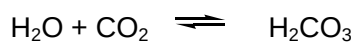
- 18** When 0.20 mol of hydrogen gas and 0.15 mol of iodine gas are heated at 723 K until equilibrium is established, the equilibrium mixture is found to contain 0.26 mol of hydrogen iodide.

The equation for the reaction is as follows.



What is the correct expression for the equilibrium constant K_c ?

- A** $\frac{2 \times 0.26}{0.20 \times 0.15}$
- B** $\frac{(2 \times 0.26)^2}{0.20 \times 0.15}$
- C** $\frac{(0.26)^2}{0.07 \times 0.02}$
- D** $\frac{(0.26)^2}{0.13 \times 0.13}$
- 19** The main buffering agent in blood plasma is the carbon dioxide/hydrogen carbonate system.



Which of the following shows how the system above acts as a buffer when OH^- ions are added to the blood plasma?

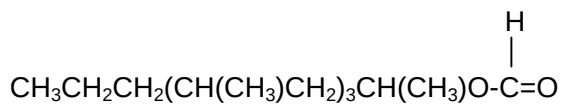
- A** $\text{CO}_2 + 2\text{OH}^- \rightarrow \text{H}_2\text{CO}_3 + \text{O}^{2-}$
- B** $\text{HCO}_3^- + \text{OH}^- \rightarrow \text{H}_2\text{O} + \text{CO}_3^{2-}$
- C** $\text{H}_2\text{CO}_3 + \text{OH}^- \rightarrow \text{HCO}_3^- + \text{H}_2\text{O}$
- D** None of the above

- 20 All the isomeric alcohols with the molecular formula $C_5H_{12}O$ are added separately to hot alkaline potassium manganate(VII).

How many of the isomers will **not** form a brown precipitate?

- A 1 B 2 C 3 D 4

- 21 The acarid mite releases Lardolure to attract other mites to a host.

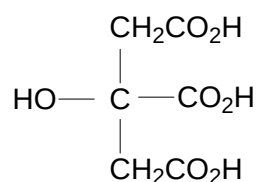


Lardolure

A simplified formula for Lardolure may be written as $RCH(CH_3)O-\overset{\overset{H}{|}}{C}=O$. This chemical can be destroyed by hydrolysis with aqueous potassium hydroxide.

What are the products obtained?

- A $RCH(CH_3)OH + CO_2$
 B $RCH(CH_3)CO_2^- K^+ + CH_3O^- K^+$
 C $RCH(CH_3)O^- K^+ + HCO_2^- K^+$
 D $RCH(CH_3)OH + HCO_2^- K^+$
- 22 Citric acid, which causes the sharp taste of lemon juice, has the following formula



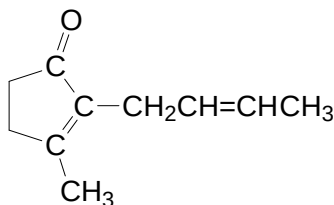
What reacts completely with 1 mol of citric acid?

- A 3 mol $PCl_5(s)$
 B 4 mol of $HCl(g)$
 C 4 mol of $NaOH(aq)$
 D 4 mol of $Na(s)$

23 What is the correct order of acid strength for the following compounds?

	Weakest	—————→	Strongest
A	Ethanoic acid	Propanol	Methanoic acid
B	Ethanoic acid	Propanoic acid	Propanol
C	2,2-dichloropropanoic acid	2-chloropropanoic acid	3-chloropropanoic acid
D	Propanoic acid	3-chloropropanoic acid	2-chloropropanoic acid

24 Jasmone is an active ingredient of jasmine. It is extracted from jasmine flowers for perfume.

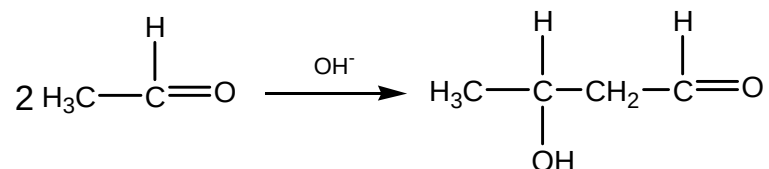


Jasmone

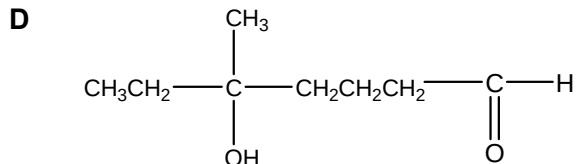
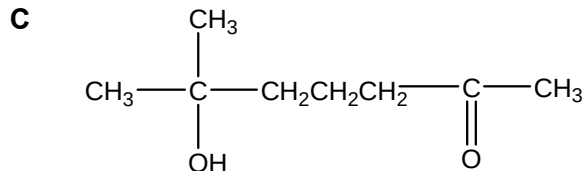
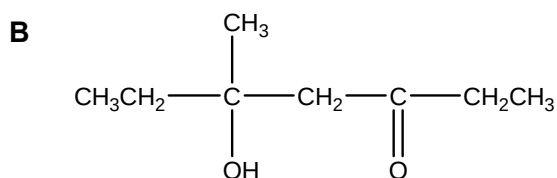
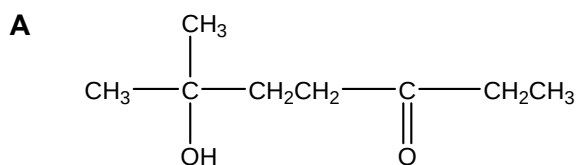
Which of the following statements is correct?

- A** It can form 1 pair of *cis-trans* isomers.
- B** All the carbon atoms in Jasmone are in the same plane.
- C** Effervescence is observed when Jasmone is heated with acidified KMnO_4 .
- D** A reddish brown precipitate is observed when hot alkaline Cu^{2+} complex solution was added to Jasmone.

- 25** Under the influence of a dilute base or acid, two molecules of an aldehyde or a ketone may combine in a process known as aldol condensation which involves the transfer of an α -hydrogen. An example of an aldol condensation between ethanal molecules is shown below.



Which of the following is a possible product formed when butanone undergoes aldol condensation?



Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

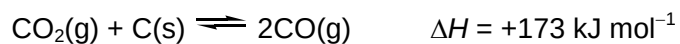
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which of the following contains the same number of particles as 16 g of oxygen atoms?

- 1** The number of molecules in 24 dm³ of Cl₂ under room conditions.
- 2** The number of ions in 1 dm³ of 1 mol dm⁻³ aqueous HCl.
- 3** The number of moles of atoms in 111.6 g of iron filings.

27



Which of the following will increase the ratio of $\frac{\text{CO}(\text{g})}{\text{CO}_2(\text{g})}$ at equilibrium?

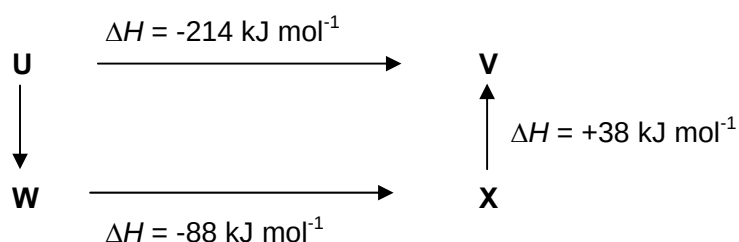
- 1** Increase in temperature
- 2** Decrease in pressure
- 3** Addition of catalyst

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

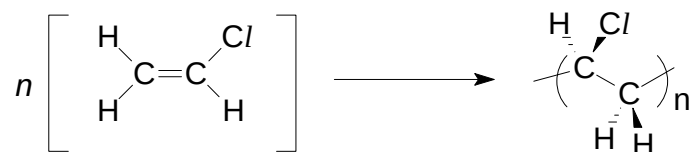
No other combination of statements is used as a correct response.

- 28** The diagram shows a set of energy changes for a set of reactions.



- 1** The enthalpy change for $\text{U} \rightarrow \text{X}$ is -176 kJ mol^{-1} .
 - 2** The enthalpy change for $\text{U} \rightarrow \text{W}$ is -164 kJ mol^{-1} .
 - 3** The enthalpy change for $\text{V} \rightarrow \text{W}$ is $+50 \text{ kJ mol}^{-1}$.
- 29** In many countries, plastic waste such as polyvinylchloride, PVC is separated from waste before it is incinerated to provide heat for power stations. This is because the products formed from its combustion destroy the ozone layer.

PVC is formed from the polymerisation of chloroethene.



Which of the following statement is true about PVC?

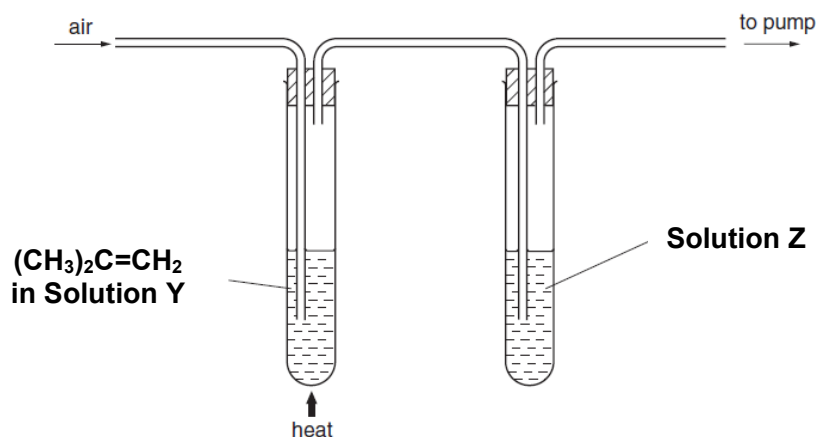
- 1** The products of combustion destroy ozone as chlorine radicals are formed.
- 2** The carbon-carbon bond in PVC is longer than that of chloroethene.
- 3** The formation of PVC is an addition reaction.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

30



Given that a precipitate is formed in solution **Z**, which of the following are the identities of the solution **Y** and **Z**?

	Solution Y	Solution Z
1	Hot acidic KMnO_4	$\text{Ca}(\text{OH})_2(\text{aq})$
2	Hot alkaline KMnO_4	2,4-dinitrophenylhydrazine
3	Hot acidic $\text{K}_2\text{Cr}_2\text{O}_7$	Aqueous alkaline iodine

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
D	B	C	C	D	A	C	B	B	C	C	A	D	C	B	B	D	C	C	A
21	22	23	24	25	26	27	28	29	30										
D	D	D	A	B	D	B	C	A	B										